Linzan Ye

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EDUCATION

Carnegie Mellon University — **Pittsburgh, Pennsylvania**

M.S. in Music and Technology — *Aug. 2025 – May 2027 (expected)*

- Research interests: online inference, language modeling, reinforcement learning, music information retrieval
- Awarded a merit-based fellowship covering 50% of tuition

University of Rochester — **Rochester, New York**

B.A. in Music (Highest Distinction) & Data Science (High Distinction) — *Aug. 2017 – May 2021*

- Cumulative GPA: 3.92 out of 4.0; Dean's List (all eligible semesters)

SKILLS

- Research: Music Information Retrieval, Audio Signal Processing, Deep Learning for Music and Audio, Sequence Models for Real-Time Music Processing, Natural Language Processing
- Programming Python, C++, Java, JavaScript/TypeScript
- Frameworks: PyTorch, Tensorflow.js, React, fluidsynth
- Languages: English (fluent), Japanese (intermediate), Chinese (native)

RELEVANT PROJECTS

Real-time AI-assisted Piano Performance — *Mar. 2025 – current*

- Trained an **LSTM**-based model for **low-latency online music generation** conditioned on button input, past history, and musical score with PyTorch.
- Deployed the system as a web-based demo using Tensorflow.js and JavaScript, achieving **<15ms end-to-end latency**.
- Designed heuristic models for **real-time** improvisation, arrangement and performance with smart tempo and velocity extrapolation.
- Preprocessed and aligned 1,068 expressive performances from the (n)ASAP dataset. Developed algorithms for XML-MIDI score-performance alignment on the (n)ASAP dataset, achieving **93.5%** average alignment rate.

Controllable Symbolic Pop Music Generation with Latent Diffusion Model — *Oct. 2024 – Jan. 2025*

- Collaborated on melody and accompaniment generation conditioned on lyrics and text prompts.
- Integrated ControlNet and MelodyT5 with a **latent diffusion model** for controllable MIDI generation.
- Trained models on NVIDIA A100 GPUs (2 days), achieving **controllable generation** over instrument, style, and key.

Masked Expressiveness – Conditioned Prediction of Piano Velocity with BERT — *Jul. 2024 – Oct. 2024*

- Implemented a **BERT**-based model for expressive piano dynamics, achieving **87.9% accuracy** within ±2 velocity bins across 32 classes.
- Designed a sequential generation algorithm that improved musical coherence in the generated performance, with measurable gains visualized through colored piano rolls.
- Enhanced a score-to-MIDI alignment algorithm, creating a demo that reconstructs full expressive performances from partial performance inputs.

Chord Sense – Stylistic Chord Progression Generation — *Mar. 2024 – Jun. 2024*

- Developed forward and backward **GPT**-based models for chord symbol generation, achieving **91.9% HITS@3**.
- Proposed and evaluated three automatic fine-tuning methods, enabling stylistic chord progression generation with limited data.
- Achieved a 5x improvement in the consistency of target progression generation across diverse harmonic contexts.

PUBLICATIONS

- Ye, L. (2024). Masked Expressiveness: Conditioned Generation of Piano Key Striking Velocity Using Masked Language Modeling. *Highlights in Science, Engineering and Technology*, *120*, 81-87
- Ye, L. (2024). Chord sense: Enhancing stylistic Chord Progression generation with fine-tuned transformers. *Applied and Computational Engineering*, 68, 359-367.

WORK EXPERIENCE

Shenzhen Mango Future Technology Co., Ltd. **Shenzhen, China**
Algorithm Engineer *Feb. 2022 – Aug. 2024*

- Led development of a **band accompaniment generation system**, reaching 3,000 daily active users. Designed a chroma-based chord classifier with 90% weighted accuracy across 54 classes using SVMs, implemented synthesizer logic, and coordinated client/server-side deployment.
- Refactored and extended a **real-time musical score synthesis and playback system** in C++, enhancing extensibility for custom instruments and audio effect plugins. Improved synthesis quality via dynamic range control, room impulse response, and custom sound fonts for realistic piano and string sounds.
- Deployed deep learning models for **expressive piano dynamic prediction** and developed a solfege-based singing module for **interactive learning**.
- Built a **conversational AI platform** for children, integrating LangChain-based evaluation systems and exploring LLM-driven generative content (music, storytelling, images). Designed a React/TypeScript admin interface with WYSIWYG for content management, review, and interactive demos.

EXTRACURRICULAR ACTIVITIES

University of Rochester **Rochester, NY**
Chamber Orchestra – Principal Clarinet *Sep. 2017 – May 2020*

- Performed in school concerts and various local venues, enhancing the art in the Rochester community.

University of Rochester **Rochester, NY**
Chamber Orchestra – Soloist *Sep. 2019 – Feb. 2020*

- The pianist winner of the University of Rochester River Campus 2019/2020 Concerto Competition. Received high praise from all judges.
- Performed John Field's Piano Concerto No. 2 mvt. 1 with the University of Rochester Chamber Orchestra at the Strong Auditorium.

SELECTED COURSEWORKS

- CMU10725 – Optimization for ML — Spring 2026
- CMU15622 – Intro to Computer Music — Spring 2026
- CMU57594 – Shaping Time in Performance — Spring 2026
- CMU10701 – Introduction to Machine Learning — Fall 2025
- CMU10734 – Foundations of Autonomous Decision Making under Uncertainty — Fall 2025
- CMU15798 – Generative AI for Music and Audio — Fall 2025
- URCSC261 – Intro to Databases — Spring 2020
- URCSC242 – Artificial Intelligence — Spring 2019
- URCSC240 – Data Mining — Fall 2019
- URCSC172 – Data Structures & Algorithms — Fall 2018
- URMUR221~224 – Music History — Fall 2018~Spring 2021
- URMUR211~212 – Music Theory — Fall 2018~Spring 2019
- URMUR114~115 – Musicianship — Spring 2019~Fall 2019
- URPA160 – Primary Piano — Fall 2018~Spring 2020